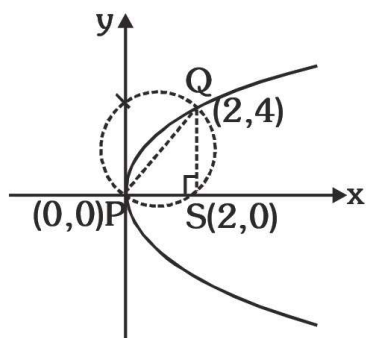


## Exercise JA

- Q1. Let S be the focus of the parabola  $y^2 = 8x$  and let PQ be the common chord of the circle  $x^2 + y^2 - 2x - 4y = 0$  and the given parabola. The area of the triangle PQS is

Ans. 4

Sol. Focus of parabola S(2,0) points of intersection of given curves : (0,0) and (2,4).

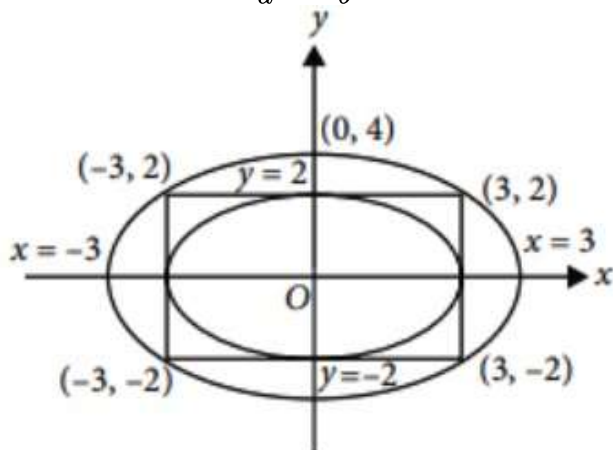


$$\text{Area } (\triangle PSQ) = \frac{1}{2} \times 2 \times 4 = 4 \text{ sq. units}$$

- Q2. The ellipse  $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$  is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse  $E_2$  passing through the point (0, 4) circumscribes the rectangle R. The length (in units) of the major axis of ellipse  $E_2$  is

Ans. 8

Sol. Let the ellipse be  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



As it is passing through (0, 4) and (3, 2).

So,  $b^2 = 16$  and  $\frac{9}{a^2} + \frac{4}{16} = 1$  or  $a^2 = 12$

So, the length of major axis  $= 2b = 8$

- Q3. Tangents are drawn to the hyperbola  $\frac{x^2}{9} - \frac{y^2}{4} = 1$ , parallel to the straight line  $2x - y = 1$ . The points of contact of the tangents on the hyperbola are

(A)  $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$  (B)  $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$  (C)  $(3\sqrt{3}, -2\sqrt{2})$  (D)  $(-3\sqrt{3}, 2\sqrt{2})$

Ans. A, B

Sol. Let parametric coordinates be  $P(3\sec\theta, 2\tan\theta)$

Equation of tangent at point P will be

$$\frac{x \sec \theta}{3} - \frac{y \tan \theta}{2} = 1$$

$\therefore$  tangent is parallel to  $2x - y = 1$

$$\Rightarrow \frac{2 \sec \theta}{3 \tan \theta} = 2 \Rightarrow \sin \theta = \frac{1}{3}$$

$\therefore$  coordinates are

$$\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \text{ and } \left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

### Paragraph

Let PQ be a focal chord of the parabola  $y^2 = 4ax$ . The tangents to the parabola at P and Q meet at a point lying on the line  $y = 2x + a$ ,  $a > 0$

\n

Q4. Length of chord PQ is

(A)  $7a$  (B)  $5a$  (C)  $2a$  (D)  $3a$

Ans. B

Sol. Length of focal chord

$$\begin{aligned} PQ &= a(t_1 - t_2)^2 \\ &= a[(t_1 + t_2)^2 - 4t_1 t_2] \\ &= a[1 + 4] = 5a \end{aligned}$$

### Paragraph

Let PQ be a focal chord of the parabola  $y^2 = 4ax$ . The tangents to the parabola at P and Q meet at a point lying on the line  $y = 2x + a$ ,  $a > 0$

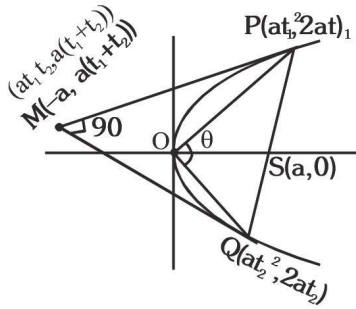
\n

Q5. If chord PQ subtends an angle  $\theta$  at the vertex of  $y^2 = 4ax$ , then  $\tan\theta =$

(A)  $\frac{2}{3}\sqrt{7}$  (B)  $\frac{-2}{3}\sqrt{7}$  (C)  $\frac{2}{3}\sqrt{5}$  (D)  $\frac{-2}{3}\sqrt{5}$

Ans. D

Sol.



chord will intersect on directrix.

$$\therefore M(-a, a(t_1 + t_2))$$

lies on  $y = 2x + a$

$$a(t_1 + t_2) = -2a + a \Rightarrow t_1 + t_2 = -1$$

$$\& \quad t_1 t_2 = -1$$

$$\tan \theta = \left( \frac{\frac{2}{t_1} - \frac{2}{t_2}}{1 + \frac{4}{t_1 t_2}} \right) = \left( \frac{2(t_2 - t_1)}{3} \right)$$

$$\therefore (t_2 - t_1)^2 = (t_2 + t_1)^2 - 4t_1 t_2 = 5$$

$$t_2 - t_1 = \pm \sqrt{5}$$

$$\therefore \tan \theta = \pm \frac{2\sqrt{5}}{3}$$

but  $\theta$  is obtuse because O is the interior point of the circle for which PQ is diameter.

$$\therefore \tan \theta = \frac{-2\sqrt{5}}{3}$$

Q6. A line  $L : y = mx + 3$  meets  $y$ -axis at  $E(0, 3)$  and the arc of the parabola  $y^2 = 16x$ ,  $0 \leq y \leq 6$  at the point  $F(x_0, y_0)$ . The tangent to the parabola at  $F(x_0, y_0)$  intersects the  $y$ -axis at  $G(0, y_1)$ . The slope  $m$  of the line  $L$  is chosen such that the area of the triangle  $EFG$  has a local maximum.

Match List I with List II and select the correct answer using the code given below the lists :

| List I                                | List II  |
|---------------------------------------|----------|
| P. $m =$                              | 1. $1/2$ |
| Q. Maximum area of $\triangle EFG$ is | 2. 4     |
| R. $y_0 =$                            | 3. 2     |
| S. $y_1 =$                            | 4. 1     |
| Codes:                                |          |

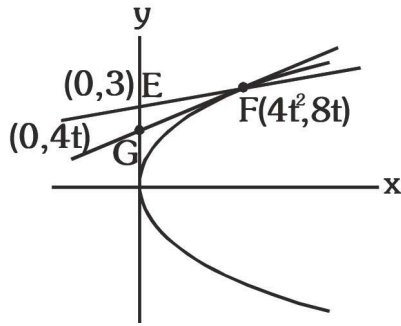
- (A) 

|   |   |   |   |
|---|---|---|---|
| P | Q | R | S |
| 4 | 1 | 2 | 3 |

 (B) 3 4 1 2 (C) 1 3 2 4 (D) 1 3 4 2

Ans. A

Sol.

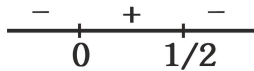


Let  $F(4t^2, 8t)$

$$\text{where } 0 \leq 8t \leq 6 \Rightarrow 0 \leq 8t \leq \frac{3}{4}$$

$$\Delta = (6t^2 - 8t^3)$$

$$\frac{d\Delta}{dt} = 12t - 24t^2 = 0 \quad \begin{cases} t = 0 (\text{minima}) \\ t = \frac{1}{2} (\text{maxima}) \end{cases}$$



$$\Rightarrow m = \frac{8t - 3}{4t^2 - 0} = \frac{4 - 3}{1} = 1$$

$$(\Delta EFG)_{\max} = \frac{6}{4} - 1 = \frac{1}{2}$$

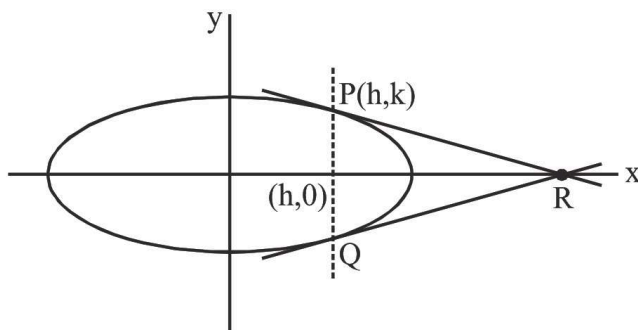
$$y_0 = 8t = 4 \text{ \& } y_1 = 4t = 2$$

Q7. A vertical line passing through the point  $(h, 0)$  intersects the ellipse  $\frac{x^2}{4} + \frac{y^2}{3} = 1$  at the point P and Q. Let the tangents to the ellipse at P and Q meet at the point R. If  $\Delta(h) =$  area of the triangle PQR,  $\Delta_1 = \max_{1/2 \leq h \leq 1} \Delta(h)$  and , then  $\Delta_2 = \min_{1/2 \leq h \leq 1} \Delta(h)$ , then

$$\frac{8}{\sqrt{5}} \Delta_1 - 8 \Delta_2 =$$

Ans. 9

Sol.



$$\text{Tangent at } P(h, k) \text{ is } \frac{xh}{4} + \frac{ky}{3} = 1$$

$$\Rightarrow R\left(\frac{4}{h}, 0\right)$$

$$\triangle PQR = k\left(\frac{4}{h} - h\right)$$

$$= \sqrt{3\left(1 - \frac{h^2}{4}\right)}\left(\frac{4}{h} - h\right)$$

which is a decreasing function in  $\left[\frac{1}{2}, 1\right]$

$$\Rightarrow \quad \triangle_1 = \sqrt{3\left(1 - \frac{1}{16}\right)}\left(8 - \frac{1}{2}\right) = \frac{45\sqrt{5}}{8}$$

$$\& \quad \triangle_2 = \sqrt{3\left(1 - \frac{1}{4}\right)}(4 - 1) = \frac{9}{2}$$

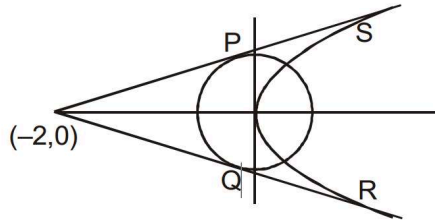
$$\frac{8}{\sqrt{5}}\triangle_1 - 8\triangle_2 = 45 - 36 = 9$$

Q8. The common tangents to the circle  $x^2 + y^2 = 2$  and the parabola  $y^2 = 8x$  touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is

- (A) 3 (B) 6 (C) 9 (D) 15

Ans. D

Sol.



$$y = mx + \frac{2}{m}$$

If it is tangent to  $x^2 + y^2 = 2$

Then,

$$\left| \frac{\frac{2}{m}}{\sqrt{1+m^2}} \right| = \sqrt{2} \Rightarrow \frac{4}{m^2(1+m^2)} = 2 \Rightarrow m \pm 1$$

Hence equation of tangent is  $y = x + 2$  &  $y = -x - 2$ .

Chord of contact PQ is  $-2x = 2$   $x = -1$

Chord of contact RS is  $y = 0 = 4(x - 2)$   $x = 2$

Hence co-ordinates of P, Q, R, S are  $(-1, 1)$ ;  $(-1, -1)$ ;  $(2, -4)$  &  $(2, 4)$

Area of trapezium is  $= \frac{1}{2}(PQ + RS) \times \text{Height} = \frac{1}{2}(10) \times 3 = 15$ .

### Paragraph

Let  $a, r, s, t$  be nonzero real numbers. Let  $P(at^2, 2at)$ ,  $Q, R(ar^2, 2ar)$  and  $S(as^2, 2as)$  be distinct points on the parabola  $y^2 = 4ax$ . Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point  $(2a, 0)$ .

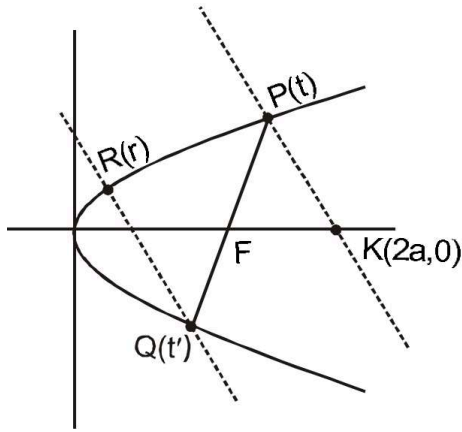
\n

Q9. The value of  $r$  is

- (A)  $-\frac{1}{t}$  (B)  $\frac{t^2 + 1}{t}$  (C)  $\frac{1}{t}$  (D)  $\frac{t^2 - 1}{t}$

Ans. A

Sol.



$$m_{PK} = m_{QR}$$

$$\frac{2at - 0}{at^2 - 2a} = \frac{2at' - 2ar}{a(t')^2 - ar^2}$$

$$\frac{t}{t^2 - 2} = \frac{t' - r}{(t')^2 - r^2}$$

$$-t' - tr^2 = -t - rt^2 - 2t' + 2r, \quad tt' = -1$$

$$t' - tr^2 = -t + 2r - rt^2$$

$$-tr^2 + r(t^2 - 2) + t' + t = 0$$

$$\lambda = \frac{(2 - t^2) \pm \sqrt{(t^2 - 2)^2 + 4(-1 + t^2)}}{-2t}$$

$$= \frac{(2 - t^2) \pm \sqrt{t^4}}{-2t} = \frac{2 - t^2 \pm t^2}{-2t}$$

$$r = -\frac{1}{t} \quad \text{It is not possible as the R \& Q will be one same.}$$

### Paragraph

Let  $a, r, s, t$  be nonzero real numbers. Let  $P(at^2, 2at)$ ,  $Q, R(ar^2, 2ar)$  and  $S(as^2, 2as)$  be distinct points on the parabola  $y^2 = 4ax$ . Suppose that  $PQ$  is the focal chord and lines  $QR$  and  $PK$  are parallel, where  $K$  is the point  $(2a, 0)$ .

\n

Q10. If  $st = 1$ , then the tangent at  $P$  and the normal at  $S$  to the parabola meet at a point whose ordinate is

$$(A) \frac{(t^2 + 1)^2}{2t^3} \quad (B) \frac{a(t^2 + 1)^2}{2t^3} \quad (C) \frac{a(t^2 + 2)^2}{t^3} \quad (D) \frac{a(t^2 + 2)^2}{t^3}$$

Ans. B

Sol.

Tangent at  $P$  is  $ty = x + at^2$

Normal at  $S$  is  $y + sx = 2as + as^2$

$$ty + x = 2a + \frac{a}{t^2}$$

$$ty = 2a = \frac{a}{t^2} - ty = at^2$$

$$2t^3y = at^4 + 2at^2 + a$$

$$y = \frac{a(t^2 + 1)^2}{2t^3}$$

Q11. Let the curve C be the mirror image of the parabola  $y^2 = 4x$  with respect to the line  $x + y + 4 = 0$ . If A and B are the points of intersection of C with the line  $y = -5$ , then the distance between A and B is

Ans. 4

Sol.

let  $P(t^2, 2t)$  be a point on the curve  $y^2 = 4x$  and  $Q(h, k)$  be it's image in  $x + y + 4 = 0$

$$\frac{h - t^2}{1} = \frac{k - 2t}{1} = -\frac{2(t^2 + 2t + 4)}{2} \Rightarrow h = -(2t + 4)$$

$$k = -(t^2 + 4)$$

$$\text{now } k = -5$$

$$\text{so } t = \pm 1$$

$$\text{hence } h = -2, -6$$

$$\text{so A, B are } (-2, -5) \text{ \& } (-6, -5)$$

$$\text{Hence } AB = 4$$

Q12. If the normals of the parabola  $y^2 = 4x$  drawn at the end point of its latus rectum are tangents to the circle  $(x-3)^2 + (y+2)^2 = r^2$ , then the value of  $r^2$  is

Ans. 2

Sol.

Equation of normals at points  $(1, \pm 2)$  are

$$y = -x + 3 \quad \& \quad y = x - 3$$

$$\Rightarrow x + y - 3 = 0 \quad \& \quad x - y - 3 = 0$$

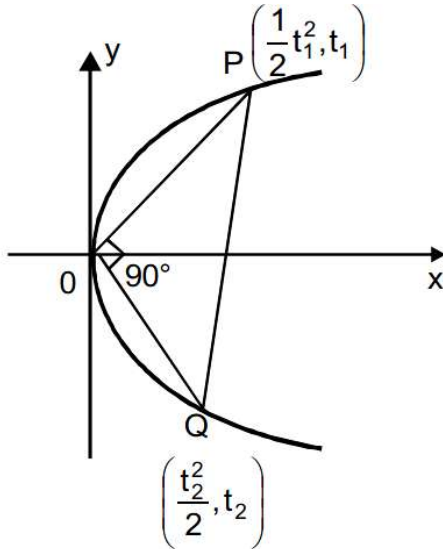
$$\text{Now } \left| \frac{3 - 2 - 3}{\sqrt{1 + 1}} \right| = r \Rightarrow r^2 = 2$$

Q13. Let P and Q be distinct points on the parabola  $y^2 = 2x$  such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle  $\Delta OPQ$  is  $3\sqrt{2}$ , then which of the following is (are) the coordinates of P ?

- (A)  $(4, 2\sqrt{2})$     (B)  $(9, 3\sqrt{2})$     (C)  $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$     (D)  $(1, \sqrt{2})$

Ans. D, A

Sol.



$$OP \perp OQ \Rightarrow t_1 t_2 = -4$$

$$\text{Now } \frac{1}{2} \cdot OP \cdot OQ = 3\sqrt{2}$$

$$\Rightarrow \left| \frac{1}{2} \cdot \sqrt{\frac{t_1^4}{4}} \cdot \sqrt{\frac{t_2^4 - 2}{4} + t_2^2} \right| = 3\sqrt{2}$$

$$\Rightarrow \frac{1}{2} \cdot 4 \cdot \sqrt{\frac{(t_1^2 + 4)(t_2^2 + 4)}{4}} = 3\sqrt{2}$$

$$\Rightarrow 4 \cdot \frac{(16 + 4(t_1^2 + t_2^2) + 16)}{16} = 9 \times 2$$

$$\Rightarrow 8 + t_1^2 + t_2^2 = 18$$

$$\Rightarrow t_1^2 + t_2^2 - 10 = 0$$

Q14. Suppose that the foci of the ellipse  $\frac{x^2}{9} + \frac{y^2}{5} = 1$  are  $(f_1, 0)$  and  $(f_2, 0)$  where  $f_1 > 0$  and  $f_2 < 0$ . Let  $P_1$  and  $P_2$  be two parabolas with a common vertex at  $(0,0)$  and with foci at  $(f_1, 0)$  and  $(2f_2, 0)$  respectively.

Let  $T_1$  be a tangent to  $P_1$  which passes through  $(2f_2, 0)$  and  $T_2$  be a tangent to  $P_2$

which passes through  $(f_1, 0)$ . If  $m_1$  is the slope of  $T_1$  and  $m_2$  is the slope of  $T_2$ , then the value of  $\left(\frac{1}{m_1^2} + m_2^2\right)$  is :

Ans. 4

Sol.  $e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{5}{9} = \frac{4}{9}$

$e = \frac{2}{3}$  focii  $(2, 0)$   $(-2, 0)$

$p_1 : y^2 = 8x,$

$y = m_1x + \frac{2}{m_1}$

$0 = -4m_1 + \frac{2}{m_1}$

$\Rightarrow 4m_1^2 = 2$

$\Rightarrow m_1 = \pm \frac{1}{\sqrt{2}}$

$p_2 : y^2 = -16x,$

$\Rightarrow y = m_2x - \frac{4}{m_2}$

$\Rightarrow 0 = 2m_2 - \frac{4}{m_2}$

$\Rightarrow 2m_2^2 = 4$

$\frac{1}{m_1^2} + m_2^2 = 2 + 2 = 4$

Q15. Let  $E_1$  and  $E_2$  be two ellipses whose centers are at the origin. The major axes of  $E_1$  and  $E_2$  lie along the x-axis and the y-axis, respectively. Let  $S$  be the circle  $x^2 + (y - 1)^2 = 2$ . The straight line  $x + y = 3$  touches the curves  $S$ ,  $E_1$  and  $E_2$  at  $P$ ,  $Q$  and  $R$ , respectively. Suppose that  $PQ = PR = \frac{2\sqrt{2}}{3}$ . If  $e_1$  and  $e_2$  are the eccentricities of  $E_1$  and  $E_2$  respectively, then the **CORRECT** expression (s) is (are) :

(A)  $e_1^2 + e_2^2 = \frac{43}{40}$  (B)  $e_1e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$  (C)  $|e_1^2 - e_2^2| = \frac{5}{8}$  (D)  $e_1e_2 = \frac{\sqrt{3}}{4}$

Ans. A, B

Sol.  $E_1 \rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$E_2 \rightarrow \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

Now as  $x + y = 3$  is a tangent

$$a^2 + b^2 = A^2 + B^2 = 9$$

Now point P is

$$x^2 + (2 - x)^2 = 2$$

$$2x^2 - 4x + 2 = 0$$

$$x = 1$$

so P is (1, 2)

points Q & R are  $\left(\frac{5}{3}, \frac{4}{3}\right)$  &  $\left(\frac{1}{3}, \frac{8}{3}\right)$

$$\text{Now } \left(\frac{5}{3}, \frac{4}{3}\right) \text{ lies on } E_1 \text{ so } \frac{25}{9a^2} + \frac{16}{9(9-a)^2} = 1$$

$$\Rightarrow 225 - 25a^2 + 16a^2 = 9a^2(9-a^2) \Rightarrow 225 - 9a^2 = 9a^2(9-a^2)$$

$$\Rightarrow 25 - a^2 = a^2(9-a^2)$$

$$\Rightarrow a^4 - 10a^2 + 25 = 0$$

$$\Rightarrow a^2 = 5 \text{ so } b^2 = 4$$

$$e_1^2 = \frac{1}{5}$$

Now  $\left(\frac{1}{3}, \frac{8}{3}\right)$  lies on E2

$$\frac{1}{A^2} + \frac{64}{(9-A^2)^2} = 9$$

$$9 - A^2 + 64A^2 = 9A^2(9-A^2)$$

$$1 + 7A^2 = A^2 = 9A^2 - A^4 \Rightarrow A^4 - 2A^2 + 1 = 0$$

$$e_2^2 = \frac{7}{8}$$

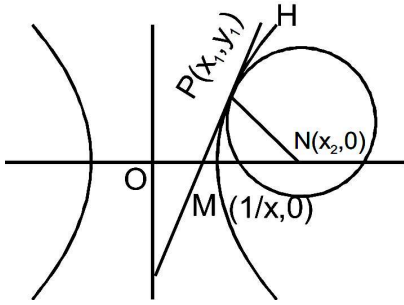
- Q16. Consider the hyperbola  $H : x^2 - y^2 = 1$  and a circle  $S$  with center  $N(x_2, 0)$ . Suppose that  $H$  and  $S$  touch each other at a point  $P(x_1, y_1)$  with  $x_1 > 1$  and  $y_1 > 0$ . The common tangent to  $H$  and  $S$  at  $P$  intersects the  $x$ -axis at point  $M$ . If  $(l, m)$  is the centroid of the triangle  $\Delta PMN$ , then the correct expression (s) is (are)

$$(A) \frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2} \text{ for } x_1 > 1 \quad (B) \frac{dm}{dx_1} = \frac{x_1}{3\left(\sqrt{x_1^2 - 1}\right)} \text{ for } x_1 > 1$$

$$(C) \frac{dl}{dx_1} = 1 + \frac{1}{3x_1^2} \text{ for } x_1 > 1 \quad (D) \frac{dm}{dy_1} = \frac{1}{3} \text{ for } y_1 > 1$$

Ans. A, B, D

Sol.



Equation tangent to H at P is  $xx_1 - yy_1 = 1$

$$l = \frac{x_1 + x_2 + \frac{1}{x_1}}{3}, \quad m = \frac{y_1}{3} = \frac{\sqrt{x_1^2 - 1}}{3}$$

$$\text{now, } \left. \frac{dy}{dx} \right|_{H \text{ at } P} = \left. \frac{dy}{dx} \right|_{S \text{ at } P} \Rightarrow \frac{x_1}{y_1} = \frac{x_2 - x_1}{y_1} \Rightarrow x_2 = 2x_1$$

$$\text{So, } l = x_1 + \frac{1}{3x_1}$$

$$\frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2}, \quad \frac{dm}{dx_1} = \frac{1}{3} \cdot \frac{x_1}{\sqrt{x_1^2 - 1}}$$

Q17. The circle  $C_1 : x^2 + y^2 = 3$ , with centre at O, intersects the parabola  $x^2 = 2y$  at the point P in the first quadrant. Let the tangent to the circle  $C_1$  at P touches other two circles  $C_2$  and  $C_3$  at  $R_2$  and  $R_3$ , respectively. Suppose  $C_2$  and  $C_3$  have equal radii  $2\sqrt{3}$  and centres  $Q_2$  and  $Q_3$ , respectively. If  $Q_2$  and  $Q_3$  lie on the y-axis, then

(A)  $Q_2Q_3 = 12$  (B)  $R_2R_3 = 4\sqrt{6}$  (C) area of the triangle  $OR_2R_3$  is  $6\sqrt{2}$

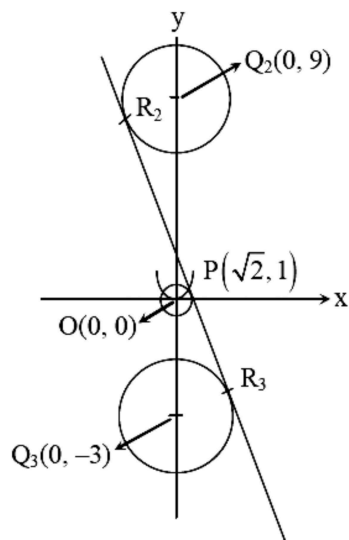
(D) area of the triangle  $PQ_2Q_3$  is  $4\sqrt{2}$

Ans. C, B, A

Sol. Equation of tangent at P  $(\sqrt{2}, 1)$  is  $\sqrt{2}x + y - 3 = 0$

If centre of  $C_2$  at  $(0, \alpha)$  and radius equal to  $2\sqrt{3}$

$$2\sqrt{3} = \left| \frac{\alpha - 3}{\sqrt{3}} \right| \Rightarrow \alpha = -3, 9$$



(A)  $Q_2Q_3 = 12$

(B)  $R_2R_3$  = length of transverse common tangent

$$= \sqrt{(Q_2Q_3)^2 - (r_1 + r_2)^2} = \sqrt{(12)^2 - (2\sqrt{3} + 2\sqrt{3})^2} = 4\sqrt{6}$$

(C) Area of  $\triangle OR_2R_3$

$$= \frac{1}{2} \times R_2R_3 \times \text{perpendicular distance of O from line}$$

$$= \frac{1}{2} \times 4\sqrt{6} \times \sqrt{3} = 6\sqrt{2}$$

(D) Area of  $\triangle PQ_2Q_3 = \frac{1}{2} \times 12 \times \sqrt{2} = 6\sqrt{2}$

Q18. Let P be the point on the parabola  $y^2 = 4x$  which is at the shortest distance from the center S of the circle  $x^2 + y^2 - 4x - 16y + 64 = 0$ . Let Q be the point on the circle dividing the line segment SP internally. Then

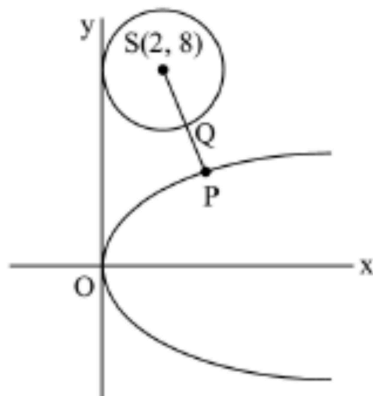
(A)  $SP = 2\sqrt{5}$     (B)  $SQ:QP = (\sqrt{5} + 1) : 2$

(C) the x-intercept of the normal to the parabola at P is 6

(D) the slope of the tangent to the circle at Q is  $\frac{1}{2}$

Ans. A, C, D

Sol.



Equation of normal of parabola

$$y + tx = 2t + t^3$$

Normal passes through S(2, 8)

$$8 + 2t = 2t + t^3$$

$$t = 2$$

Hence P (4, 4) and SQ = radius = 2

### Paragraph

Let  $F_1(x_1, 0)$  and  $F_2(x_2, 0)$  for  $x_1 < 0$  and  $x_2 > 0$ , be the foci of the ellipse  $\frac{x^2}{9} + \frac{y^2}{8} = 1$ .

Suppose a parabola having vertex at the origin and focus at  $F_2$  intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

\n

Q19. The orthocentre of the triangle  $F_1MN$  is

- (A)  $\left(-\frac{9}{10}, 0\right)$  (B)  $\left(\frac{2}{3}, 0\right)$  (C)  $\left(\frac{9}{10}, 0\right)$  (D)  $\left(\frac{2}{3}, \sqrt{6}\right)$

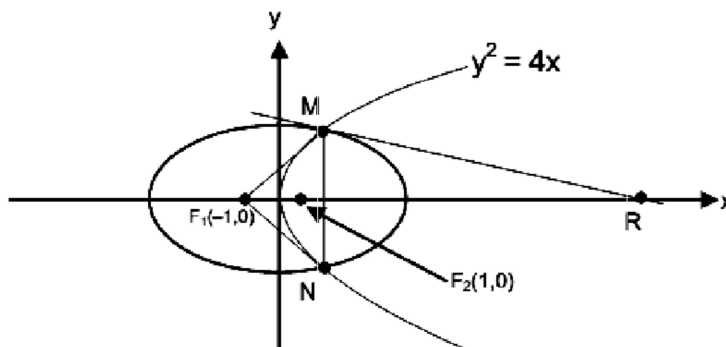
Ans. A

Sol. Ellipse :  $\frac{x^2}{9} + \frac{y^2}{8} = 1$  .....(1)

foci of ellipse are  $(\pm 1, 0)$

Equation of parabola having vertex (0, 0) and focus (1, 0) is  $y^2 = 4x$

.....(2)



from equation (1) & (2)  $\frac{x^2}{9} + \frac{4x}{8} = 1 \Rightarrow 2x^2 + 9x - 18 = 0 \Rightarrow x = \frac{3}{2}, -6$  (rejected)

$\therefore M\left(\frac{3}{2}, \sqrt{6}\right)$  and  $N\left(\frac{3}{2}, -\sqrt{6}\right)$

Equation of altitude from vertex  $M\left(\frac{3}{2}, \sqrt{6}\right)$  is  $(y - \sqrt{6}) = \frac{5}{2\sqrt{6}}\left(x - \frac{3}{2}\right)$

put  $y = 0$  we get  $\Rightarrow \frac{-12}{5} = x - \frac{3}{2} \Rightarrow x = \frac{-9}{10}$

$\Rightarrow$  orthocenter of  $\triangle F_1 MN$  is  $\left(\frac{-9}{10}, 0\right)$

### Paragraph

Let  $F_1(x_1, 0)$  and  $F_2(x_2, 0)$  for  $x_1 < 0$  and  $x_2 > 0$ , be the foci of the ellipse  $\frac{x^2}{9} + \frac{y^2}{8} = 1$ .

Suppose a parabola having vertex at the origin and focus at  $F_2$  intersects the ellipse at point  $M$  in the first quadrant and at point  $N$  in the fourth quadrant.

\n

Q20. If the tangents to the ellipse at  $M$  and  $N$  meet at  $R$  and the normal to the parabola at  $M$  meets the  $x$ -axis at  $Q$ , then the ratio of area of the triangle  $MQR$  to area of the quadrilateral  $MF_1NF_2$  is

(A) 3 : 4    (B) 4 : 5    (C) 5 : 8    (D) 2 : 3

Ans. C

Sol. Equation of tangent at point  $M\left(\frac{3}{2}, \sqrt{6}\right)$  to the ellipse is  $\frac{x(3/2)}{9} + \frac{y\sqrt{6}}{8} = 1$

Equation of normal to the parabola at point  $M\left(\frac{3}{2}, \sqrt{6}\right)$  is  $(y - \sqrt{6})\left(x - \frac{3}{2}\right)$

put  $y = 0 \Rightarrow Q$  is  $\left(\frac{7}{2}, 0\right)$

$$\text{Now } \frac{\text{Area of } \triangle MQR}{\text{Area of quadrilateral } MF_1NF_2} = \frac{\frac{1}{2} \cdot \frac{5}{2} \cdot \sqrt{6}}{\frac{1}{2} \cdot 2.2\sqrt{6}} = \frac{5}{8}$$

Q21. If a chord, which is not a tangent, of the parabola  $y^2 = 16x$  has the equation  $2x + y = p$ , and midpoint  $(h, k)$ , then which of the following is(are) possible value(s) of  $p$ ,  $h$  and  $k$ ?

- (A)  $p = -2, h = 2, k = -4$     (B)  $p = 2, h = 3, k = -4$     (C)  $p = -1, h = 1, k = -3$   
 (D)  $p = 5, h = 4, k = -3$

Ans. B

Sol.  $T = S_1$   
 $k^2 - 16h = ky - 8(x + h)$   
 $ky - 8x = k^2 - 8h$   
 compare with  $2x + y = p$   
 $-\frac{8}{2} = \frac{k}{1} = \frac{k^2 - 8h}{p}$   
 Check option.

Q22. If  $2x - y + 1 = 0$  is a tangent to the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$ , then which of the following CANNOT be sides of a right angled triangle?

- (A)  $a, 4, 2$     (B)  $a, 4, 1$     (C)  $2a, 8, 1$     (D)  $2a, 4, 1$

Ans. A, B, C

Sol. For  $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$   
 Condition of tangency for  $y = mx + c$  is  
 $c^2 = a^2m^2 - b^2$   
 $1 = a^2(4) - 16$   
 $4a^2 = 17$   
 $2a = \pm\sqrt{17}$   
 (A)  $a^2 \neq 4^2 + 2^2$     (B)  $a^2 \neq 4^2 + 1^2$     (C)  $(2a)^2 \neq 8^2 + 1^2$     (D)  $(2a)^2 = 4^2 + 1^2$

### Paragraph

Q.46 to 48 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contain conics, equations of tangents to the conics and points of contact, respectively.

Columns 1, 2 and 3 contain conics, equations of tangents to the conics and points of contact, respectively.

| Column-1 | Column-2 | Column-3 |
|----------|----------|----------|
|----------|----------|----------|



\n

Q24. For a  $= \sqrt{2}$ , if a tangent is drawn to suitable conic (Column 1) at the point of contact  $(-1, 1)$ , then which of the following options is the only CORRECT combination for obtaining its equation?

(A) (I) (i) (P)      (B) (III) (i) (P)      (C) (II) (ii) (Q)      (D) (I) (ii) (Q)

Ans. D

Sol.  $(-1, 1)$  will satisfy curve I for  $a = \sqrt{2}$  I  $\rightarrow$  (ii)  $\rightarrow$  Q

## Paragraph

Q.46 to 48 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contain conics, equations of tangents to the conics and points of contact, respectively.

| Column-1                  | Column-2                                                     | Column-3                                                                         |
|---------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| (I) $x^2 + y^2 = a^2$     | (i) $my = m^2x + a$                                          | (p) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$                                   |
| (II) $x^2 + a^2y^2 = a^2$ | $y = mx + a\sqrt{m^2 + 1}$                                   | (Q) $\left(\frac{-ma}{\sqrt{m^2 + 1}}, \frac{a}{\sqrt{m^2 + 1}}\right)$          |
| (III) $y^2 = 4ax$         | (iii) $y = mx + \frac{\sqrt{a^2m^2 - 1}}{\sqrt{a^2m^2 - 1}}$ | (R) $\left(\frac{-a^2m}{\sqrt{a^2m^2 + 1}}, \frac{1}{\sqrt{a^2m^2 + 1}}\right)$  |
| (IV) $x^2 - a^2y^2 = a^2$ | (iv) $y = mx + \frac{\sqrt{a^2m^2 + 1}}{\sqrt{a^2m^2 + 1}}$  | (S) $\left(\frac{-a^2m}{\sqrt{a^2m^2 - 1}}, \frac{-1}{\sqrt{a^2m^2 - 1}}\right)$ |

\\n\\n

$\backslash n$

\n

\n

Q25. If a tangent to a suitable conic (Column 1) is found to be  $y = x + 8$  and its point of contact is  $(8, 16)$ , then which of the following options is the only CORRECT combination?

(A) (III) (i) (P)      (B) (I) (ii) (Q)      (C) (II) (iv) (R)      (D) (III) (ii) (Q)

Ans. A

Sol.  $m = 1, \quad C = 8$   
 for  $y^2 = 4ax \Rightarrow (8, 16)$  will satisfy for  $a = 8$   
 $4 - mx + \frac{8}{m}$

$$y = x + 8$$

III  $\rightarrow$  (i)  $\rightarrow$  P

Q26. Consider two straight lines, each of which is tangent to both the circle  $x^2 + y^2 = \frac{1}{2}$  and the parabola  $y^2 = 4x$ . Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin O(0, 0) and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is  $\sqrt{2}$ , then which of the following statement(s) is (are) TRUE ?

(A) For the ellipse, the eccentricity is  $\frac{1}{\sqrt{2}}$  and the length of the latus rectum is 1

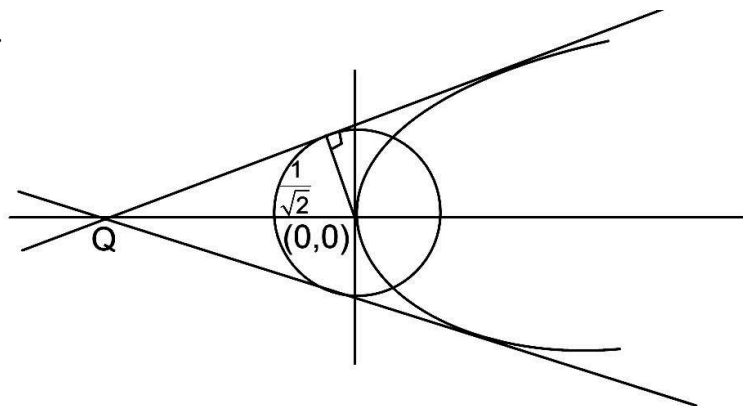
(B) For the ellipse, the eccentricity is  $\frac{1}{2}$  and the length of the latus rectum is  $\frac{1}{2}$

The area of the region bounded by the ellipse between the lines  $x = \frac{1}{\sqrt{2}}$  and  $x = 1$   
 (C) is  $\frac{1}{4\sqrt{2}}(\pi - 2)$

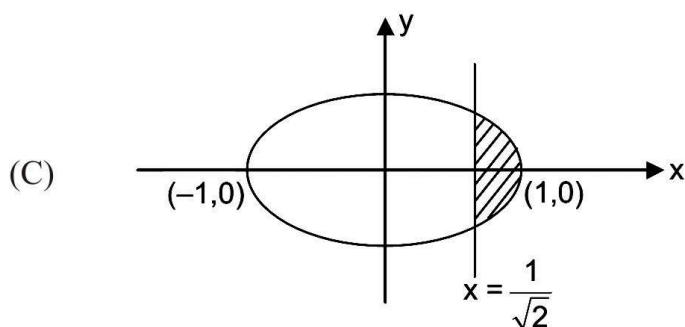
The area of the region bounded by the ellipse between the lines  $x = \frac{1}{\sqrt{2}}$  and  $x = 1$   
 (D) is  $\frac{1}{16}(\pi - 2)$

Ans. A, C

Sol.



$$\left| \frac{0+0+\frac{1}{m}}{\sqrt{1+m^2}} \right| = \frac{1}{\sqrt{2}} \Rightarrow m^4 + m^2 - 2 = 0 \Rightarrow m = \pm 1 \text{ Equation of common tangents are } y = x + 1 \text{ \& } y = x - 1$$



$$\text{Area} = 2 \int_{1/\sqrt{2}}^1 \frac{1}{\sqrt{2}} \cdot \sqrt{1-x^2} dx = \sqrt{2} \left[ \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right]_{1/\sqrt{2}}^1 = \sqrt{2} \left[ \frac{x}{4} - \left( \frac{1}{4} + \frac{\pi}{8} \right) \right] = \sqrt{2} \left( \frac{\pi}{8} - \frac{1}{4} \right)$$

- Q27. Let  $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , where  $a > b > 0$ , be a hyperbola in the  $xy$ -plane whose conjugate axis  $LM$  subtends an angle of  $60^\circ$  at one of its vertices  $N$ . Let the area of the triangle  $LMN$  be  $4\sqrt{3}$ .

LIST-I

P. The length of the conjugate axis of  $H$  is

Q. The eccentricity of  $H$  is

R. The distance between the foci of  $H$  is

S. The length of the latus rectum of  $H$  is

The correct option is:

LIST-II

1. 8

2.  $\frac{4}{\sqrt{3}}$

3.  $\frac{2}{\sqrt{3}}$

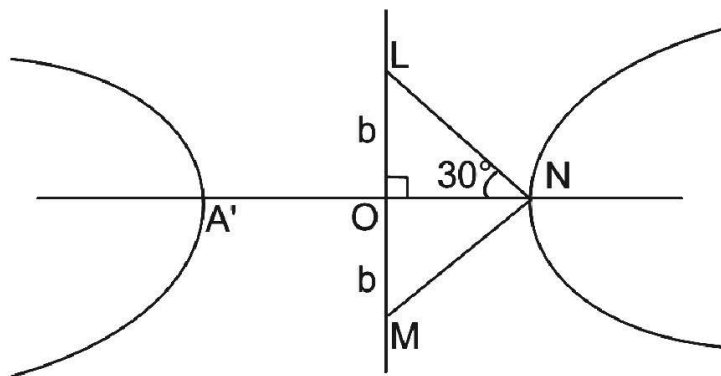
4. 4

(A)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$  (B)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C)  $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$  (D)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

Ans. 5f0aba2690e4b0a1523b63bd

Sol.



$$\text{Area of LMN} = 4\sqrt{3}$$

$$\frac{1}{2}(2b)(\sqrt{3}b) = 4\sqrt{3} \Rightarrow b^2 = 4 \Rightarrow b = 2 \quad \text{P} \quad 2b = 4$$

$$\text{Here } \frac{a}{b} = \cot 30^\circ \Rightarrow a = \sqrt{3}b \Rightarrow a = 2\sqrt{3}$$

$$b^2 = a^2(e^2 - 1)$$

$$4 = 12(e^2 - 1)$$

$$e^2 + \frac{1}{3} = \frac{4}{3} \Rightarrow e = \frac{2}{\sqrt{3}} \quad \text{and} \quad 2ae = 2 \times 2\sqrt{3} \times \frac{2}{\sqrt{3}} = 8$$

$$\text{and length of latus rectum } \frac{2b^2}{a} = \frac{2 \times 4}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

### Paragraph

This section contains TWO (02) List-Match sets.

Let the circle  $C_1: x^2 + y^2 = 9$  and  $C_2: (x - 3)^2 + (y - 4)^2 = 16$  intersect at the points X and Y. Suppose that another circle  $C_3: (x - h)^2 + (y - k)^2 = r^2$  satisfies the following conditions.

(i) Centre of  $C_3$  is collinear with the centres of  $C_1$  &  $C_2$ ,

(ii)  $C_1$  &  $C_2$  both lie inside  $C_3$  and

(iii)  $C_3$  touches  $C_1$  at M and  $C_2$  at N

Let the line through X and Y intersect  $C_3$  at Z and W and let a common tangent of  $C_1$  &  $C_3$  be a tangent to the parabola  $x^2 = 8\alpha y$

There are some expressions given in the list-I, whose values are given in list-II below :

List I

List II

(I)  $2h + k$

(P) 6

(II)  $\frac{\text{Length of } ZW}{\text{Length of } XY}$

(Q)  $\sqrt{6}$

(III)  $\frac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$

(R)  $\frac{5}{4}$

(IV)  $\alpha$

(S)  $\frac{21}{5}$

(T)  $2\sqrt{6}$

(U)  $\frac{10}{3}$

Q28. Which of the following is the only incorrect combination :

- (A) (III), (R)    (B) (IV), (S)    (C) (I), (P)    (D) (IV), (U)

Ans. B

Sol. -

### Paragraph

This section contains TWO (02) List-Match sets.

Let the circle  $C_1: x^2 + y^2 = 9$  and  $C_2: (x - 3)^2 + (y - 4)^2 = 16$  intersect at the points X and Y. Suppose that another circle  $C_3: (x - h)^2 + (y - k)^2 = r^2$  satisfies the following conditions.

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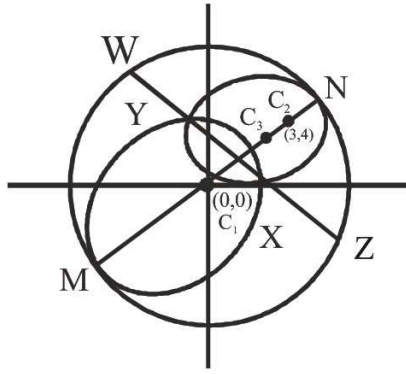
| List I                                                                    | List II            |
|---------------------------------------------------------------------------|--------------------|
| (I) $2h + k$                                                              | (P) 6              |
| (II) $\frac{\text{Length of } ZW}{\text{Length of } XY}$                  | (Q) $\sqrt{6}$     |
| (III) $\frac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$ | (R) $\frac{5}{4}$  |
| (IV) $\alpha$                                                             | (S) $\frac{21}{5}$ |
|                                                                           | (T) $2\sqrt{6}$    |
|                                                                           | (U) $\frac{10}{3}$ |

Q29. Which of the following is the only correct combination :

- (A) (I), (U)    (B) (II), (Q)    (C) (I), (S)    (D) (II), (T)

Ans. B

Sol.  $C_1: x^2 + y^2 = 9$      $C_1(0, 0)$      $r_1 = 3$   
 $C_2: (x - 3)^2 + (y - 4)^2 = 16$      $C_2: (3, 4)$      $r_2 = 4$   
 $C_3: (x - h)^2 + (y - k)^2 = r^2$      $C_3: (h, k)$      $r_3 = r$   
 $MN = MC_1 + C_1C_2 + C_2N$   
 $MN = r_1 + \sqrt{9 + 16} + r_2$   
 $MN = 3 + 5 + 4 = 12 = 2r \Rightarrow r = 6$   
 $C_3, C_1$  and  $C_2$  are collinear  
 $C_1(0, 0) C_2(3, 4)$



$$\frac{4}{3} = \frac{y}{x} \quad \Rightarrow y = \frac{4x}{3}$$

$$C_3(h, k) \quad k = \frac{4h}{3}$$

$$C_3 : \left( h, \frac{4h}{3} \right)$$

$$C_1 C_3 = MC_3 - C_1 M$$

$$\sqrt{h^2} = \frac{16h^2}{9} = 6 - 3 = 3$$

$$\frac{5|h|}{3} = 3 \Rightarrow |h| = \frac{9}{5} \Rightarrow h = \frac{9}{5}$$

$$k = \frac{4h}{3} = \frac{4}{3} \times \frac{9}{5} = \frac{12}{5}$$

$$(I) \quad 2h + k = \frac{18}{5} + \frac{12}{5} = \frac{30}{5} = 6$$

(II) Equation of line ZW

$$C_1 - C_2 = 0$$

$$x^2 + y^2 - 9 - x^2 - 9 + 6x - y^2 + 8y - 16 + 16 = 0$$

$$6x + 8y = 18$$

$$3x + 4y = 9$$

Distance of ZW from  $C_1(0, 0)$

$$\left| \frac{-9}{\sqrt{25}} \right| = \frac{9}{5}$$

$$\text{Length XY} = 2\sqrt{3^2 - \left(\frac{9}{5}\right)^2}$$

$$= 2\sqrt{9 - \frac{81}{25}} = \sqrt{\frac{225 - 81}{25}}$$

$$= 2\sqrt{\frac{144}{25}} = \frac{12 \times 2}{5} = \frac{24}{5}$$

$$C_3 : \left( \frac{9}{5}, \frac{12}{5} \right)$$

Distance of  $C_3$  from ZW

$$\left| \frac{3 \times \frac{9}{5} + 4 \times \frac{12}{5} - 9}{5} \right|$$

$$\left| \frac{\frac{27}{5} + \frac{48}{5} - 9}{5} \right| = \left| \frac{27 + 48 - 45}{25} \right| = \left| \frac{30}{25} \right| = \frac{6}{5}$$

$$\text{Length of WZ} = \sqrt{36 - \frac{36}{25}}$$

$$= 12\sqrt{\frac{24}{25}} = \frac{12}{5} \times 2\sqrt{6} = \frac{24}{5}\sqrt{6}$$

$$\frac{\text{Length of ZW}}{\text{Length of XY}} = \frac{\frac{24\sqrt{6}}{5}}{\frac{12}{5} \times 2}$$

$$\text{(III) Area of } \triangle MZN = \frac{1}{2}(MN) \left( \frac{1}{2}ZW \right)$$

$$= \frac{1}{4} \times 12 \times \frac{24}{5} \times \sqrt{6}$$

$$= \frac{72\sqrt{6}}{5}$$

$$\text{Area of } \triangle ZMW = \frac{1}{2}(ZW) (\text{OM} + \text{Distance of } C_1 \text{ from ZW})$$

$$= \frac{1}{2} \times \frac{24\sqrt{6}}{5} \times \left( 3 + \frac{9}{5} \right)$$

$$= \frac{12\sqrt{6}}{5} \times \left( \frac{24}{5} \right) = \frac{288\sqrt{6}}{25}$$

$$\frac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}} = \frac{\frac{72\sqrt{6}}{5}}{\frac{288\sqrt{6}}{25}}$$

$$= \frac{72\sqrt{6}}{5} \times \frac{25}{288\sqrt{6}} = \frac{5}{4}$$

$$\text{(IV) Slope of tangent to } C_1 \text{ at M} = \frac{-3}{4}$$

Equation of tangent

$$y = mx - 3\sqrt{1+m^2}$$

$$y = \frac{-3x}{4} - 3\sqrt{1 + \frac{9}{16}}$$

$$y = \frac{-3x}{4} - \frac{3 \times 5}{4}$$

$$4y = -3x - 15$$

$$4y + 3x + 15 = 0 \dots\dots\dots \text{(i)}$$

$$T \Rightarrow x = my + \frac{2a}{m}$$

$$3x + 4y + 15 = 0$$

$$mx - m^2y - 2a = 0$$

$$\frac{3}{m} = \frac{-4}{m^2} = \frac{-15}{2\alpha}$$

$$3m^2 = -4m$$

$$3m^2 + 4m = 0$$

$$m(3m + 4) = 0$$

$$m = \frac{-4}{3}$$

$$\frac{3 \times 3}{-4} = \frac{-15}{2\alpha} \Rightarrow \alpha = \frac{10}{3}$$

Q30. Define the collections  $\{E_1, E_2, E_3, \dots\}$  of ellipses and  $\{R_1, R_2, R_3, \dots\}$  of rectangles as follows :

$$E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1;$$

$R_1$  : rectangle of largest area, with sides parallel to the axes, inscribed in  $E_1$ ;

$E_n$  : ellipse  $\frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1$  of largest area inscribed in  $R_{n-1}$ ,  $n > 1$ ;

$R_n$  : rectangle of largest area, with sides parallel to the axes, inscribed in  $E_n$ ,  $n > 1$ .

Then which of the following options is/are correct ?

(A) The eccentricities of  $E_{18}$  and  $E_{19}$  are NOT equal

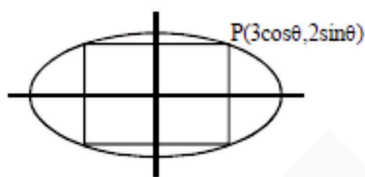
(B)  $\sum_{n=1}^N (\text{area of } R_n) < 24$ , for each positive integer  $N$

(C) The length of latus rectum of  $E_9$  is  $\frac{1}{6}$

(D) The distance of a focus from the centre in  $E_9$  is  $\frac{\sqrt{5}}{32}$

Ans. -

Sol.



Area of  $R_1 = 3\sin 2\theta$  ; for this to be maximum

$$\Rightarrow \theta = \frac{\pi}{4} \Rightarrow \left( \frac{3}{\sqrt{2}}, \frac{2}{\sqrt{2}} \right)$$

Hence for subsequent areas of rectangles  $R_n$  to be maximum the coordinates will be in GP with common

$$\text{ratio } r = \frac{1}{\sqrt{2}} \Rightarrow a_n = \frac{3}{(\sqrt{2})^{n-1}}; b_n = \frac{2}{(\sqrt{2})^{n-1}}$$

Eccentricity of all the ellipses will be same

$$\text{Distance of a focus from the centre in } E_9 = a_9 e_9 = \sqrt{a_9^2 - b_9^2} = \frac{\sqrt{5}}{16}$$

$$\text{Length of latus rectum of } E_9 = \frac{2b_9^2}{a_9} = \frac{1}{6}$$

$$\therefore \sum_{n=1}^{\infty} \text{Area of } R_n \dots 12 + \frac{12}{2} + \frac{12}{4} + \dots \dots \infty = 24$$

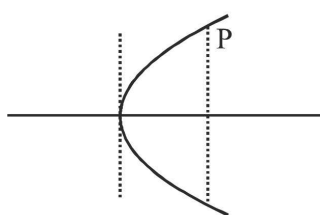
$$\Rightarrow \sum_{n=1}^N (\text{area of } R_n) < 24, \text{ for each positive integer } N$$

Q31. Let  $a$ ,  $b$  and  $l$  be positive real numbers. Suppose  $P$  is an end point of the latus rectum of the parabola  $y^2 = 4lx$ , and suppose the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  passes through the point  $P$ . If the tangents to the parabola and the ellipse at the point  $P$  are perpendicular to each other, then the eccentricity of the ellipse is.

- (A)  $\frac{1}{\sqrt{2}}$  (B)  $\frac{1}{2}$  (C)  $\frac{1}{3}$  (D)  $\frac{2}{5}$

Ans. A

Sol.



$P(d, 2\lambda)$

slope of tangent at  $P$

$$m_1 = 1$$

for ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{dy}{dx} = -\frac{xb^2}{ya^2}$$

$$m_{(\lambda, 2\lambda)} = -1$$

$$\frac{b^2}{a^2} = 2$$

$\therefore$  ellipse is vertical.

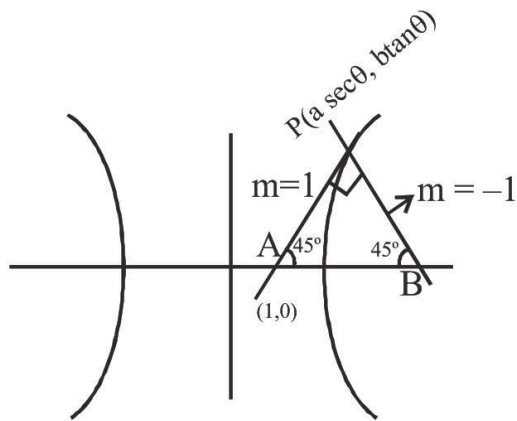
$$e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

Q32. Let  $a$  and  $b$  be positive real numbers such that  $a > 1$  and  $b < a$ . Let  $P$  be a point in the first quadrant that lies on the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ . Suppose the tangent to the hyperbola at  $P$  passes through the point  $(1, 0)$ , and suppose the normal to the hyperbola at  $P$  cuts off equal intercepts on the coordinate axes. Let  $D$  denote the area of the triangle formed by the tangent at  $P$ , the normal at  $P$  and the  $x$ -axis. If  $e$  denotes the eccentricity of the hyperbola, then which of the following statements is/are TRUE?

- (A)  $1 < e < \sqrt{2}$     (B)  $\sqrt{2} < e < 2$     (C)  $\triangle = a^4$     (D)  $\triangle = b^4$

Ans. A, D

Sol.



equation of tangent  $\Rightarrow T = 0$

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

passing through  $(1, 0)$

$$a = \sec \theta$$

$$m = \frac{b \sec \theta}{a \tan \theta} = 1$$

$$b = \tan \theta$$

$$p(a^2, b^2) \text{ or } p(\sec^2 \theta, \tan^2 \theta)$$

$$b^2 = a^2(e^2 - 1)$$

$$\tan^2 \theta = \sec^2 \theta (e^2 - 1)$$

$$e^2 - 1 = \sin^2 \theta$$

$$0 < \sin^2 \theta < 1$$

$$1 < 1 + \sin^2 \theta < 2$$

$$1 < e^2 < \sqrt{2}$$

$$1 < e < \sqrt{2}$$

$$\text{Area of } \triangle PAB = \frac{1}{2}(AP)(PB)$$

$$= \frac{1}{2}(AP)^2 \quad \{\because PA = PB\}$$

$$= \frac{1}{2}\{\sec^2 \theta - 1)^2 + (\tan^2 \theta - 0)^2\}$$

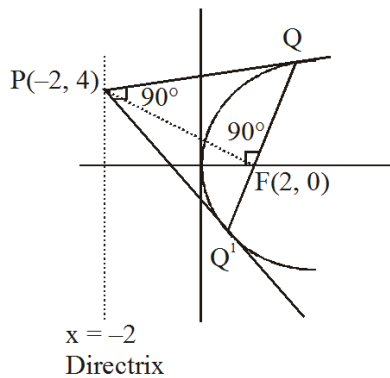
$$= \tan^4 \theta = b^4$$

Q33. Let E denote the parabola  $y^2 = 8x$ . Let  $P = (-2, 4)$ , and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E. Let F be the focus of E. Then which of the following statements is (are) TRUE ?

- (A) The triangle PFQ is a right-angled triangle  
 (B) The triangle QPQ' is a right-angled triangle  
 (C) The distance between P and F is  $5\sqrt{2}$  (D) F lies on the line joining Q and Q'

Ans. A, B, D

Sol.



P lies on the directrix of the parabola.

$\Rightarrow$  QQ' is focal chord.

$\Rightarrow \angle QPQ' = 90^\circ$  (Directrix is also director circle of the parabola)

$\Rightarrow \angle PFQ = 90^\circ$  (Portion of tangent intercepted between point of contact and directrix subtends right angle at focus.)

$\Rightarrow PF = \sqrt{4^2 + 4^2} = 4\sqrt{2}$

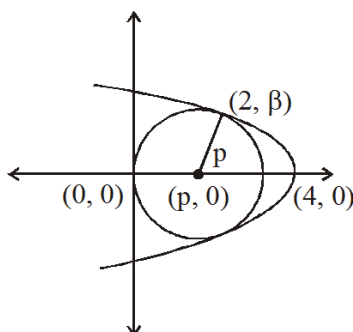
As (ABD)

Q34. Consider the region  $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$ . Let F be the family of all circles that are contained in R and have centers on the x-axis. Let C be the circle that has largest radius among the circles in F. Let  $(\alpha, \beta)$  be a point where the circle C meets the curve  $y^2 = 4 - x$ .

The radius of the circle C is \_\_\_\_\_.

Ans. 1.49, 1.51

Sol.



$$C : x^2 + y^2 - 2px = 0$$

$$P : y^2 = 4 - x$$

By solving both the curves

$$x^2 - (2p + 1)x + 4 = 0$$

$D = 0$  (Both curves touch each other)

$$(2p + 1)^2 - 42 = 0$$

$$(2p + 5)(2p - 3) = 0 \Rightarrow p = 3/2 = \text{radius of circle}$$

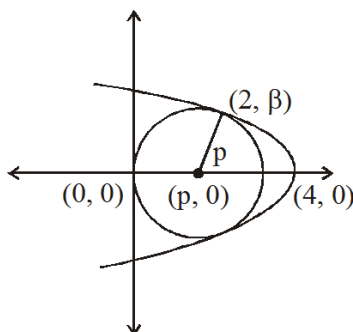
$$\text{for } p = \frac{3}{2} \Rightarrow x^2 - 4x + 4 = 0 \Rightarrow x = 2 = a$$

- Q35. Consider the region  $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$ . Let  $F$  be the family of all circles that are contained in  $R$  and have centers on the  $x$ -axis. Let  $C$  be the circle that has largest radius among the circles in  $F$ . Let  $(\alpha, \beta)$  be a point where the circle  $C$  meets the curve  $y^2 = 4 - x$ .

The value of  $\alpha$  is \_\_\_\_ .

Ans. 1.95, 2.05

Sol.



$$C : x^2 + y^2 - 2px = 0$$

$$P : y^2 = 4 - x$$

By solving both the curves

$$x^2 - (2p + 1)x + 4 = 0$$

$D = 0$  (Both curves touch each other)

$$(2p + 1)^2 - 42 = 0$$

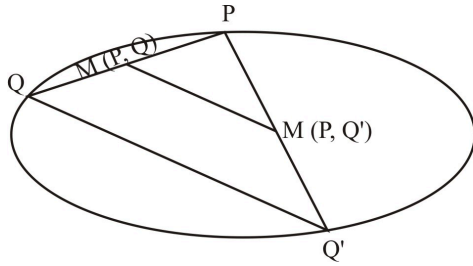
$$(2p + 5)(2p - 3) = 0 \Rightarrow p = 3/2 = \text{radius of circle}$$

$$\text{for } p = \frac{3}{2} \Rightarrow x^2 - 4x + 4 = 0 \Rightarrow x = 2 = a$$

- Q36. Let  $E$  be the ellipse  $\frac{x^2}{16} + \frac{y^2}{9} = 1$ . For any three distinct points  $P$ ,  $Q$  and  $Q'$  On  $E$ , let  $M(P, Q)$  be the mid-point of the line segment joining  $P$  and  $Q$ , and  $M(P, Q')$  be the mid-point of the line segment joining  $P$  and  $Q'$ . Then the maximum possible value of the distance between  $M(P, Q)$  and  $M(P, Q')$ , as  $P$ ,  $Q$  and  $Q'$  vary on  $E$ , is\_\_\_\_\_.

Ans. 4

Sol.



We know from triangle geometry distance between  $M(P, Q)$  and  $M(P, Q') = \frac{1}{2} QQ'$

hence distance between  $M(P, Q)$  and  $M(P, Q')$  will be maximum if  $QQ'$  is maximum and in given ellipse maximum value of  $QQ'$  will be  $2a$

then maximum distance between  $M(P, Q)$  and  $M(P, Q') = \frac{1}{2} (2a) = \frac{8}{2} = 4$

- Q37. Consider the parabola  $y^2 = 4x$ . Let  $S$  be the focus of the parabola. A pair of tangents drawn to the parabola from the point  $P = (-2, 1)$  meet the parabola at  $P_1$  and  $P_2$ . Let  $Q_1$  and  $Q_2$  be points on the lines  $SP_1$  and  $SP_2$  respectively such that  $PQ_1$  is perpendicular to  $SP_1$  and  $PQ_2$  is perpendicular to  $SP_2$ . Then, which of the following is/are TRUE ?

(A)  $SQ_1 = 2$     (B)  $Q_1Q_2 = \frac{3\sqrt{10}}{5}$     (C)  $PQ_1 = 3$     (D)  $SQ_2 = 1$

Ans. B, C, D

Sol. Let equation of tangent with slope 'm' be

$$T : y = mx + \frac{1}{m}$$

T : passes through  $(-2, 1)$  so

$$1 = -2m + \frac{1}{m}$$

$$\Rightarrow m = -1 \text{ or } m = \frac{1}{2}$$

Points are given by  $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

So, one point will be  $(1, -2)$  and  $(4, 4)$

$$P_1S : 4x - 3y - 4 = 0$$

$$P_2S : x - 1 = 0$$

$$PQ_1 = \left| \frac{4(-2) - 3(1) - 4}{5} \right| = 3$$

$$SP = \sqrt{10}; PQ_2 = 3; SQ_1 = 1 = SQ_2$$

$$\frac{1}{2} \left( \frac{Q_1Q_2}{2} \right) \times \sqrt{10} = \frac{1}{2} \times 3 \times 1 \quad (\text{comparing Areas})$$

$$\Rightarrow Q_1Q_2 = \frac{2 \times 3}{\sqrt{10}} = \frac{3\sqrt{10}}{5}$$

Q38. Consider the ellipse  $\frac{x^2}{4} + \frac{y^2}{3} = 1$ .

Let  $H(\alpha, 0)$ ,  $0 < \alpha < 2$ , be a point. A straight line drawn through  $H$  parallel to the  $y$ -axis crosses the ellipse and its auxiliary circle at points  $E$  and  $F$  respectively, in the first quadrant. The tangent to the ellipse at the point  $E$  intersects the positive  $x$ -axis at a point  $G$ . Suppose the straight line joining  $F$  and the origin makes an angle  $\phi$  with the positive  $x$ -axis.

| List -I                                                         | List -II                         |
|-----------------------------------------------------------------|----------------------------------|
| (I) If $\phi = \frac{\pi}{4}$ , then the area of the triangle   | (P) $\frac{(\sqrt{3} - 1)^4}{8}$ |
| (II) If $\phi = \frac{\pi}{3}$ , then the area of the triangle  | (Q) 1                            |
| (III) If $\phi = \frac{\pi}{6}$ , then the area of the triangle | (R) $\frac{3}{4}$                |
| (IV) If $\phi = \frac{\pi}{12}$ , then the area of the triangle | (S) $\frac{1}{2\sqrt{3}}$        |
|                                                                 | (T) $\frac{3\sqrt{3}}{2}$        |

The correct option is :

(A) (I)  $\rightarrow$  (R); (II)  $\rightarrow$  (S); (III)  $\rightarrow$  (Q); (IV)  $\rightarrow$  (P)

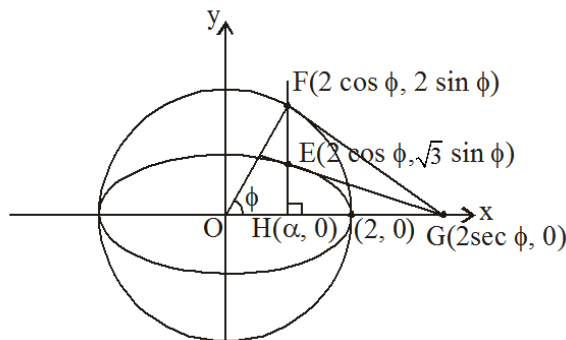
(B) (I)  $\rightarrow$  (R); (II)  $\rightarrow$  (T); (III)  $\rightarrow$  (S); (IV)  $\rightarrow$  (P)

(C) (I)  $\rightarrow$  (Q); (II)  $\rightarrow$  (T); (III)  $\rightarrow$  (S); (IV)  $\rightarrow$  (P)

(D) (I)  $\rightarrow$  (Q); (II)  $\rightarrow$  (S); (III)  $\rightarrow$  (Q); (IV)  $\rightarrow$  (P)

Ans. 631342d6f69e6e3404af2223

Sol.



$$\alpha \equiv 2 \cos \phi$$

Tangent at  $E(2 \cos \phi, \sqrt{3} \sin \phi)$  to ellipse  $\frac{x^2}{4} + \frac{y^2}{3} = 1$

$$\text{i.e. } \frac{x \cos \phi}{2} + \frac{y \sin \phi}{\sqrt{3}} = 1 \quad G(2 \sec \phi, 0)$$

$$\text{Area of triangle } FGH = \frac{1}{2}(2 \sec \phi - 2 \cos \phi)2 \sin \phi$$

$$\Delta = 2 \sin^2 \phi \cdot \tan \phi$$

I.  $\phi = \frac{\pi}{4}, \Delta = 1 \rightarrow (Q)$

II.  $\phi = \frac{\pi}{3}, \Delta = 2 \cdot \left(\frac{\sqrt{3}}{2}\right)^2 \cdot \sqrt{3} = \frac{3\sqrt{3}}{2} \rightarrow (T)$

III. If  $\phi = \frac{\pi}{6}, \Delta = 2 \cdot \left(\frac{1}{2}\right)^2 \cdot \frac{1}{\sqrt{3}} = \frac{1}{2\sqrt{3}} \rightarrow (S)$

IV. If  $\phi = \frac{\pi}{12}, \Delta = \left(1 - \frac{\sqrt{3}}{2}\right) \cdot (2 - \sqrt{3}) = \frac{(2 - \sqrt{3})^2}{2} \rightarrow (P)$

- Q39. Consider the hyperbola  $\frac{x^2}{100} - \frac{y^2}{64} = 1$  with foci at S and  $S_1$ , where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let  $\angle SPS_1 = \alpha$ , with  $\alpha < \frac{\pi}{2}$ . The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line  $S_1P$  at  $P_1$ . Let d be the distance of P from the straight line  $SP_1$ , and  $\beta = S_1P$ . Then the greatest integer less than or equal to  $\frac{\beta\delta}{9} \sin \frac{\alpha}{2}$  is \_\_\_\_.

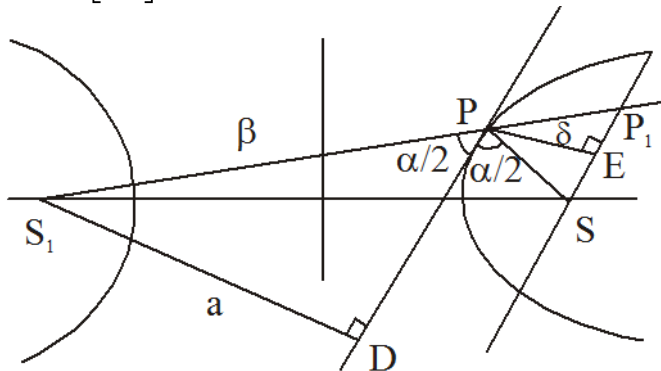
Ans. 7

Sol. Let  $S_1D = a$  in  $S_1PD$   $\sin\left(\frac{\alpha}{2}\right) = \frac{a}{\beta}$

Now  $\frac{1}{9}\beta\delta \sin\left(\frac{\alpha}{2}\right) = \frac{1}{9}\beta\delta \cdot \frac{a}{\beta} = \frac{a\delta}{9} = \frac{b^2}{9} = \frac{64}{9}$

we know product of perpendiculars from foci upon any tangent to Hyperbola =  $b^2$   
( $a\delta = b^2$ )

Hence  $\left[\frac{64}{9}\right] = 7$



- Q40. Let P be a point on the parabola  $y^2 = 4ax$ , where  $a > 0$ . The normal to the parabola at P meets the x-axis at a point Q. The area of the triangle PFQ, where F is the focus of the parabola, is 120. If the slope m of the normal and a are both positive integers, then the pair (a, m) is :

- (A) (2,3) (B) (1,3) (C) (2,4) (D) (3,4)

Ans. A

Sol.  $y = mx - 2am - am^3$

Q  $(2a + am^2, 0)$

$$\Delta PFQ = \frac{1}{2} \times h \times FQ = 120$$

$$\frac{1}{2} |-2am| (a + am^2) = 120$$

$$a^2 |m(1 + m^2)| = 120$$

Now check options

$$a = 2 \quad m = 3$$

Q41. Let  $T_1$  and  $T_2$  be two distinct common tangents to the ellipse  $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$  and the parabola  $P : y^2 = 12x$ . Suppose that the tangent  $T_1$  touches P and E at the points  $A_1$  and  $A_2$ , respectively and the tangent  $T_2$  touches P and E at the points  $A_4$  and  $A_3$ , respectively. Then which of the following statements is(are) true?

- (A) The area of the quadrilateral  $A_1 A_2 A_3 A_4$  is 35 square units
- (B) The area of the quadrilateral  $A_1 A_2 A_3 A_4$  is 36 square units
- (C) The tangents  $T_1$  and  $T_2$  meet the x-axis at the point  $(-3, 0)$
- (D) The tangents  $T_1$  and  $T_2$  meet the x-axis at the point  $(-6, 0)$

Ans. A, C

Sol.  $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ , Tangent :  $y = m_1 x \pm \sqrt{6m_1^2 + 3}$

$P : y^2 = 12x$ , Tangent :  $y = m_2 x + \frac{3}{m_2}$

For common tangent  $m = m_1 = m_2$

$$\pm \sqrt{6m^2 + 3} = \frac{3}{m}$$

$$\Rightarrow m = \pm 1$$

$\Rightarrow$  equation of common tangents are  $y = x + 3$  and  $y = -x - 3$

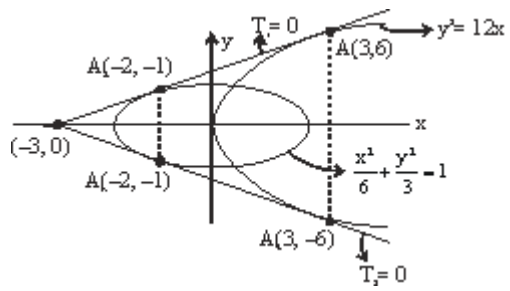
point of contact for parabola is  $\left( \frac{a}{m^2}, \frac{2a}{m} \right)$

$$\Rightarrow A_1 \equiv (3, 6), A_4 (3, -6)$$

Let  $A_2 (x_1, y_1) \Rightarrow$  tangent to E is  $\frac{xx_1}{6} + \frac{yy_1}{3} = 1$

$$A_2 (x_1, y_1) = A_2 (-2, 1)$$

$$A_3 \text{ is mirror image of } A_2 \text{ in x-axis} \Rightarrow A_3 (-2, -1)$$



Intersection point of  $T_1 = 0$  and  $T_2 = 0$  is  $(-3, 0)$

Area of quadrilateral  $A_1A_2A_3A_4 = \frac{1}{2}(12 + 2) \times 5 = 35$  square units

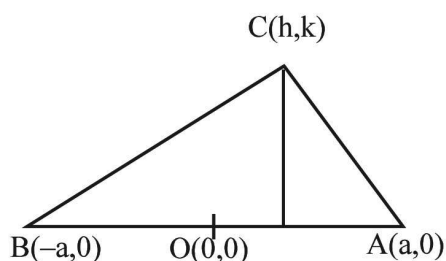
## Exercise C1

Q1. If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is :

- (A) a circle    (B) a parabola    (C) an ellipse    (D) a hyperbola

Ans. B

Sol.



$$\tan A = \frac{k}{a-h}; \tan B = \frac{k}{a+h}$$

$\tan A + \tan B = 'C'$  (a constant)

$$\frac{k}{a-h} + \frac{k}{a+h} = C \Rightarrow \frac{k(a+h) + k(a-h)}{a^2 - h^2} = C$$

$$\therefore a^2 - h^2 = \frac{2a}{C} \cdot k \Rightarrow \text{locus of } (h, k) \text{ will be}$$

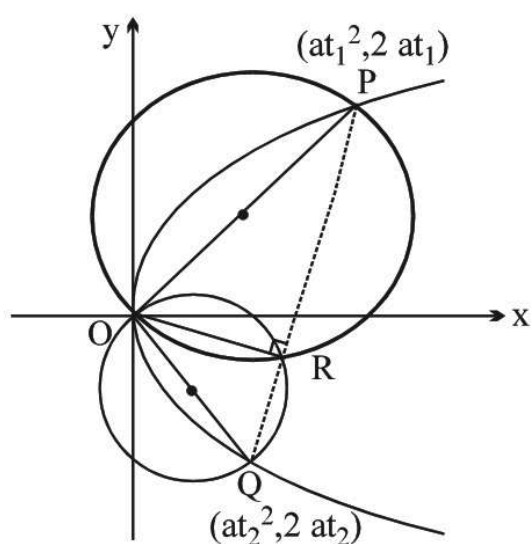
$$x^2 = a^2 - \frac{2a}{C} \cdot y \Rightarrow \text{A parabola Ans.}$$

Q2. Through the vertex O of the parabola,  $y^2 = 4ax$  two chords OP & OQ are drawn and the circles on OP & OQ as diameters intersect in R. If  $\theta_1, \theta_2$  &  $\phi$  are the angles made with the axis by the tangents at P & Q on the parabola & by OR then the value of,  $\cot \theta_1 + \cot \theta_2 =$

- (A)  $-2 \tan \phi$     (B)  $-2 \tan (\pi - \phi)$     (C) 0    (D)  $2 \cot \phi$

Ans. A

Sol.



Slope of tangent at P is  $\frac{1}{t_1}$

and at Q =  $\frac{1}{t_2}$

$$\Rightarrow \cot\theta_1 = t_1 \text{ and } \cot\theta_2 = t_2$$

$$\text{Slope of PQ} = \frac{2}{t_1 + t_2}$$

$$\Rightarrow \text{Slope of OR is } \left( -\frac{t_1 + t_2}{2} \right) = \tan\phi$$

(Note angle in a semicircle is  $90^\circ$ )

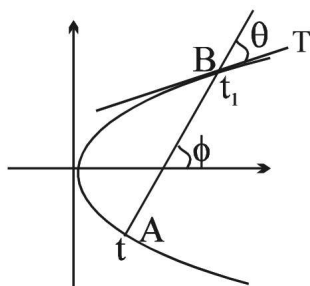
$$\Rightarrow \tan\phi = -\frac{1}{2}(\cot\theta_1 + \cot\theta_2) \Rightarrow \cot\theta_1 + \cot\theta_2 = -2\tan\phi$$

Q3. If a normal to a parabola  $y^2 = 4ax$  makes an angle  $\phi$  with its axis, then it will cut the curve again at an angle

(A)  $\tan^{-1}(2\tan\phi)$  (B)  $\tan^{-1}\left(\frac{1}{2}\tan\phi\right)$  (C)  $\cot^{-1}\left(\frac{1}{2}\tan\phi\right)$  (D)  $\cot^{-1}\left(\frac{1}{2}\tan\phi\right)$

Ans. B

Sol.



equation of normal at  $t$  :  $y + tx = 2at + at^3$

$$\therefore m_N \text{ at A} = -t = \tan\phi$$

$$t = -\tan\phi = m_1$$

Now tangent at B  $t_1y = x + at_1^2$  with  $\frac{1}{\tan\theta}$

$$\text{also } t_1 = -t - \frac{2}{t}$$

$$\therefore \tan\theta = \left| \frac{\frac{t}{t_1} + t}{1 - \frac{t}{t_1}} \right| = \left| \frac{1 + tt^2}{t - t_1} \right| = \left| \frac{1 + t^2}{2\left(t + \frac{1}{t}\right)} \right| = \left| \frac{\sec^2\phi \cdot \tan\phi}{2(\sec^2\phi)} \right|$$

$$[\text{As } t t_1 = -t^2 - 2]$$

$$\text{Hence } \tan\theta = \frac{\tan\phi}{2} \Rightarrow \theta = \tan^{-1}\left(\frac{\tan\phi}{2}\right)$$

Q4. If  $x + iy = \sqrt{\phi + i\Psi}$  where  $i = \sqrt{-1}$  and  $\phi$  and  $\Psi$  are non zero real parameters then  $\phi =$  constant and  $\Psi =$  constant, represents two systems of rectangular hyperbola which intersect at an angle of

- (A)  $\frac{\pi}{6}$  (B)  $\frac{\pi}{3}$  (C)  $\frac{\pi}{4}$  (D)  $\frac{\pi}{2}$

Ans. D

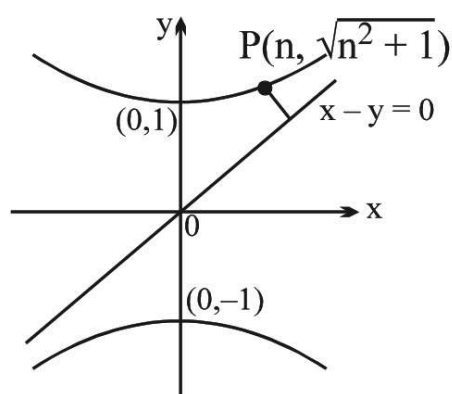
Sol.  $x^2 - y^2 + 2xyi = \phi + \psi$ ;  $x^2 - y^2 = \phi$  and  $xy = \psi$ ; which intersects at  $\frac{\pi}{2} \Rightarrow$  (D)

Q5. For each positive integer  $n$ , consider the point  $P$  with abscissa  $n$  on the curve  $y^2 - x^2 = 1$ . If  $d_n$  represents the shortest distance from the point  $P$  to the line  $y = x$  then  $\lim_{n \rightarrow \infty} (n \cdot d_n)$  has the value equal to

- (A)  $\frac{1}{2\sqrt{2}}$  (B)  $\frac{1}{2}$  (C)  $\frac{1}{\sqrt{2}}$  (D) 0

Ans. A

Sol.



Curve is rectangular hyperbola.

$$\text{perpendicular distance, } d_n = \left| \frac{n - \sqrt{n^2 + 1}}{\sqrt{2}} \right|$$

$$\lim_{n \rightarrow \infty} (n \cdot d_n) = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}} \left| \left( \sqrt{n^2 + 1} - n \right) \right|$$

$$= \lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}} \left| \frac{1}{\sqrt{n^2 + 1} + n} \right| = \frac{1}{2\sqrt{2}} \quad \text{Ans.}$$

Q6. The locus of the mid-point of the line segment joining the focus to a moving point on the parabola  $y^2 = 4ax$  is another parabola with directrix

- (A)  $x = -a$  (B)  $x = -a/2$  (C)  $x = 0$  (D)  $x = a/2$

Ans. C

Sol.  $\alpha = \frac{at^2 + a}{2}, \beta = \frac{2at + 0}{2} \Rightarrow 2\alpha = at^2 + a, at = \beta$

$$\therefore 2\alpha = a \cdot \frac{\beta}{a^2} + a \quad \text{or} \quad 2a\alpha = \beta^2 = a^2$$

$$\therefore \text{The locus is } y^2 = \frac{4a}{2} \left( x - \frac{a}{2} \right) = 4b(x - b) \left( b = \frac{a}{2} \right)$$

$$\therefore \text{Directrix is } (x - b) + b = 0 \quad \text{or} \quad x = 0$$

Q7. The equation of the common tangent to the curves  $y^2 = 8x$  and  $xy = -1$  is

- (A)  $3y = 9x + 2$  (B)  $y = 2x + 1$  (C)  $2y = x + 8$  (D)  $y = x + 2$

Ans. D

Sol. The given curves are

$$y^2 = 8x \quad \dots (1)$$

$$\text{and} \quad xy = -1 \quad \dots (2)$$

If  $m$  is the slope of tangent to (1), then equation of tangent is

$$y = \frac{mx + 2}{m}$$

If this tangent is also a tangent to (2), then

$$x \left( mx + \frac{2}{m} \right) = -1$$

$$\Rightarrow \quad mx^2 + \frac{2}{m}x + 1 = 0$$

$$\therefore \quad m^2x^2 + 2x + m = 0$$

We should get repeated roots for this equation (conditions of tangency)

$$\Rightarrow \quad D = 0$$

$$\therefore \quad (2)^2 - 4m^2 \cdot m = 0$$

$$\Rightarrow \quad m^3 = 1$$

$$\Rightarrow \quad m = 1$$

Hence required tangent is  $y = x + 2$ .

Q8. The slope of the focal chords of the parabola  $y^2 = 16x$  which are tangents to the circle  $(x - 6)^2 + y^2 = 2$  are

- (A)  $\pm 2$  (B)  $-1/2, 2$  (C)  $\pm 1$  (D)  $-2, 1/2$

Ans. C

Sol. Let slope is  $m$

$y - 0 = m(x - 4)$  focal chord

$$mx - y - 4m = 0$$

tangent to circle

$$(x - 6)^2 + y^2 = 2$$

$$\left| \frac{6m - 4m}{\sqrt{1 + m^2}} \right| = \sqrt{2}$$

$$2m = \sqrt{2}\sqrt{1 + m^2}$$

$$4m^2 = 2 + 2m^2$$

$$m^2 = 1$$

$$m = \pm 1$$

Q9. The angle between the tangents drawn from the point (1, 4) to the parabola  $y^2 = 4x$  is

(A)  $\frac{\pi}{2}$     (B)  $\frac{\pi}{3}$     (C)  $\frac{\pi}{4}$     (D)  $\frac{\pi}{6}$

Ans. B

Sol. tangent

$$y = mx + \frac{a}{m}$$

$$a = 1 \text{ passing } (1, 4)$$

$$4 = m + \frac{1}{m}$$

$$m^2 - 4m + 1 = 0 \quad \begin{matrix} \nearrow m_1 \\ \searrow m_2 \end{matrix}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\sqrt{16 - 4}}{1 + 1} \right| = \sqrt{3}$$

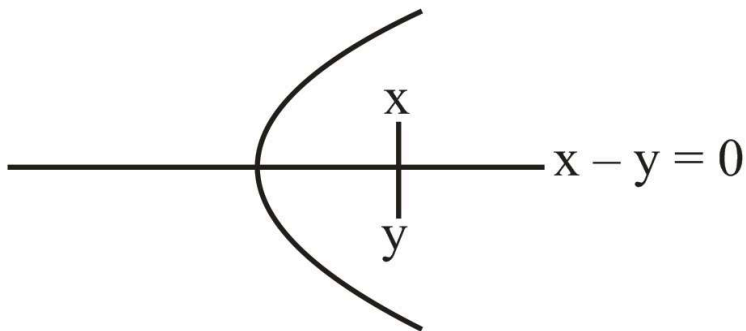
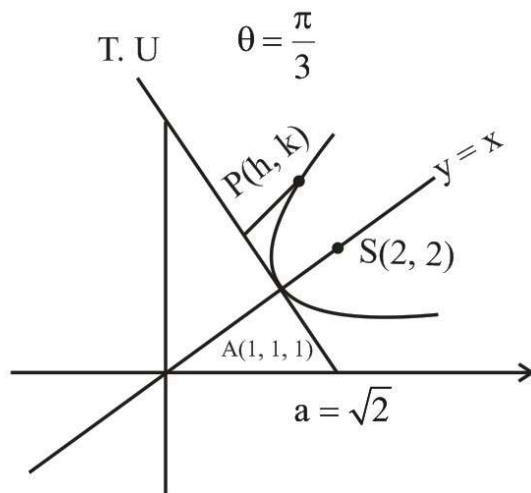
Q10. The axis of parabola is along the line  $y = x$  and the distance of vertex from origin is  $\sqrt{2}$  and that of origin from its focus is  $2\sqrt{2}$ . If vertex and focus both lie in the 1st quadrant, then the equation of the parabola is

(A)  $(x + y)^2 = (x - y - 2)$     (B)  $(x - y)^2 = (x + y - 2)$     (C)  $(x - y)^2 = 4(x + y - 2)$

(D)  $(x - y)^2 = 8(x + y - 2)$

Ans. D

Sol.



T.V (tangent at vertex)

$$y - 1 = (x - 1)$$

$$x + y = 2$$

$$\left(\frac{h-k}{\sqrt{2}}\right)^2 = 4\sqrt{2}\left(\frac{-h+k-2}{\sqrt{2}}\right)$$

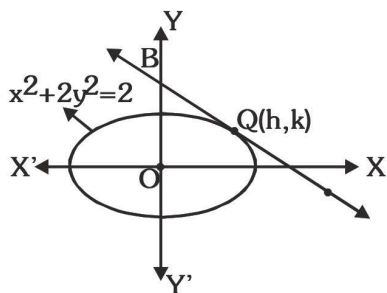
$$(x-y)^2 = (x+y-2)$$

Q11. The locus of the middle point of the intercept of the tangents drawn from an external point to the ellipse  $x^2 + 2y^2 = 2$ , between the coordinates axes, is

(A)  $\frac{1}{x^2} + \frac{1}{2y^2} = 1$     (B)  $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$     (C)  $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$     (D)  $\frac{1}{2x^2} + \frac{1}{y^2} = 1$

Ans. C

Sol.



Let the point of contact be

$$R \equiv (\sqrt{2} \cos \theta, \sin \theta)$$

Equation of tangent AB is

$$\frac{x}{\sqrt{2}} \cos \theta + y \sin \theta = 1$$

$$\Rightarrow \cos \theta = \frac{1}{h\sqrt{2}}, \sin \theta = \frac{1}{2k} \Rightarrow \frac{1}{2h^2} + \frac{1}{4k^2} = 1$$

$$\therefore \text{Required locus is } \frac{1}{2x^2} + \frac{1}{4y^2} = 1$$

Trick : The locus of mid-points of the portion of tangents to the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  intercepted between axes is  $a^2y^2 + b^2x^2 = 4x^2y^2$

$$\text{i.e., } \frac{a^2}{4x^2} + \frac{b^2}{4y^2} = 1 \text{ or } \frac{1}{2x^2} + \frac{1}{4y^2} = 1$$

Q12. The minimum area of triangle formed by the tangent to the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and coordinate axes is

(A)  $ab$  sq. units      (B)  $\frac{a^2 + b^2}{2}$  sq. units      (C)  $\frac{(a + b)^2}{2}$  sq. units

(D)  $\frac{a^2 + ab + b^2}{3}$  sq. units

Ans. A

Sol. Equation of tangent at  $(a \cos \theta, b \sin \theta)$  is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$P = \left( \frac{a}{\cos \theta}, 0 \right)$$

$$Q = \left( 0, \frac{b}{\sin \theta} \right)$$

$$\text{Area of OPQ} = \frac{1}{2} \left| \left( \frac{a}{\cos \theta} \right) \left( \frac{b}{\sin \theta} \right) \right| = \frac{ab}{|\sin 2\theta|}$$

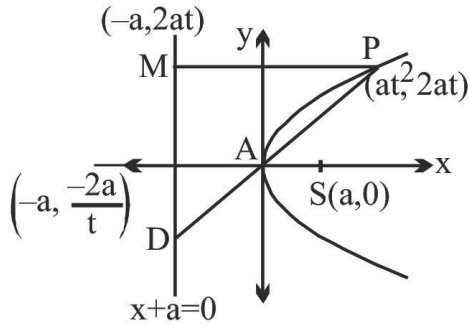
$$\therefore (\text{Area})_{\min} = ab$$

Q13. P is a point on the parabola  $y^2 = 4ax$  ( $a > 0$ ) whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x-axis at a point whose co-ordinates are :

(A)  $(-3a, 0)$       (B)  $(-a, 0)$       (C)  $(-2a, 0)$       (D)  $(a, 0)$

Ans. D, A

Sol.



$$\text{Circle : } (x + a)^2 + (y - 2at)^2 = (2at)^2$$

$$\text{from } y = 0, \quad x^2 + 2ax - 3a^2 = 0 \Rightarrow x = a \text{ or } -3a$$

Q14. The parabola  $x = y^2 + ay + b$  intersect the parabola  $x^2 = y$  at  $(1, 1)$  at right angle. Which of the following is/are correct?

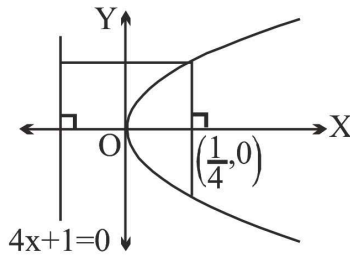
(A)  $a = 4, b = -4$     (B)  $a = 2, b = -2$

(C) Equation of the director circle for the parabola  $x = y^2 + ay + b$  is  $4x + 1 = 0$ .

(D) Area enclosed by the parabola  $x = y^2 + ay + b$  and its latus rectum is  $\frac{1}{2}$ .

Ans. D, C

Sol.



$$\text{We have } x = y^2 + ay + b \quad \dots(1) \quad \text{and} \quad x^2 = y \quad \dots(2)$$

As (1) passing through  $(1, 1)$

$$\Rightarrow 1 = 1 + a + b \Rightarrow a + b = 0$$

$$\text{Now } 1 = 2y \frac{dy}{dx} + a \frac{dy}{dx}$$

$$\left[ \frac{dy}{dx} \right]_{(1,1)} = \frac{1}{2+a} \text{ and for } y = x^2, \quad \frac{dy}{dx} = 2x \Rightarrow \left[ \frac{dy}{dx} \right]_{(1,1)} = 2$$

$$\text{Hence Required area} = 2 \int_0^{\frac{1}{4}} \sqrt{x} dx = (2) \left( \frac{2}{3} \right) \left( x^{\frac{3}{2}} \right)_0^{\frac{1}{4}} = \frac{1}{6} \text{ sq. units.} \quad \text{Ans.} \Rightarrow (D)$$

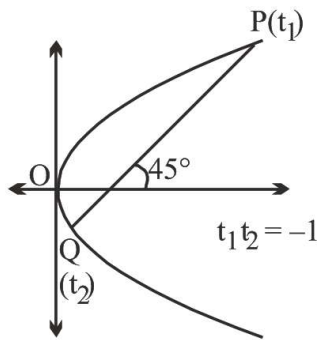
$$\text{Equation of directrix is } x = \frac{-1}{4} \Rightarrow 4x + 1 = 0 \quad \text{Ans.} \Rightarrow (C).$$

Q15. A circle 'S' is described on the focal chord of the parabola  $y^2 = 4x$  as diameter. If the focal chord is inclined at an angle of  $45^\circ$  with axis of  $x$ , then which of the following is/are true?

- (A) Radius of the circle is 4. (B) Centre of the circle is (3, 2).  
 (C) The line  $x + 1 = 0$  touches the circle.  
 (D) The circle  $x^2 + y^2 + 2x - 6y + 3 = 0$  is orthogonal to 'S'.

Ans. A, B, C

Sol.



Circle on PQ as diameter is  $(x - t_1^2)(x - t_2^2) + (y - 2t_1)(y - 2t_2) = 0$

$$x^2 + y^2 - (t_1^2 + t_2^2)x - 2(t_1 + t_2)x - 2(t_1 + t_2)y + t_1^2 t_2^2 + 4t_1 t_2 = 0$$

But  $\frac{t_1 + t_2}{2} = 1 \Rightarrow t_1 + t_2 = 2$  and  $t_1 t_2 = -1$

Hence equation of the circle is

$$x^2 + y^2 - ((t_1 + t_2)^2 - 2t_1 t_2)x - 2(t_1 + t_2)y + t_1^2 t_2^2 + 4t_1 t_2 = 0$$

$$x^2 + y^2 - 6x - 4y - 3 = 0 \quad \text{.....(1) with centre (3, 2) and radius} = 4$$

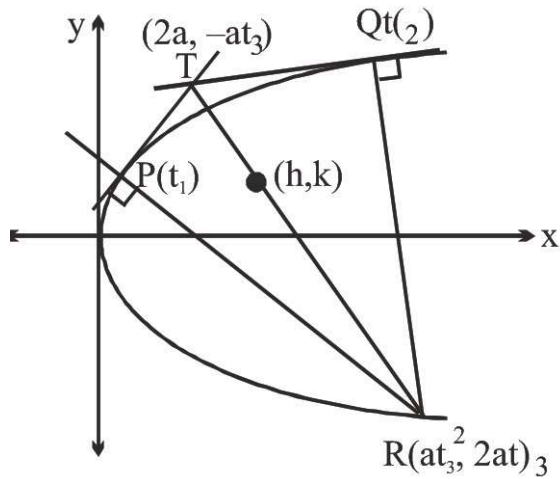
(1) is not orthogonal to  $x^2 + y^2 + 2x - 6y + 3 = 0$

Q16. TP and TQ are tangents to parabola  $y^2 = 4x$  and normals at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing  $\Delta TPQ$  is a parabola whose

- (A) vertex is (1, 0) (B) foot of directrix is  $(\frac{7}{8}, 0)$  (C) length of latus-rectum is  $\frac{1}{4}$   
 (D) focus is  $(\frac{9}{8}, 0)$

Ans. D, B, A

Sol.



Here  $a = 1$

$$t_1 t_2 = 2$$

$$t_1 + t_2 + t_3 = 0$$

We have  $2h = t_3^2 + 2$  ....(1)

$$2k = t_3 \quad \text{....(2)}$$

$$\therefore 2h = 4k^2 + 2$$

$$\therefore 2y^2 = x - 1$$

$$y^2 = \frac{1}{2} (x - 1) \quad (\text{Parabola})$$

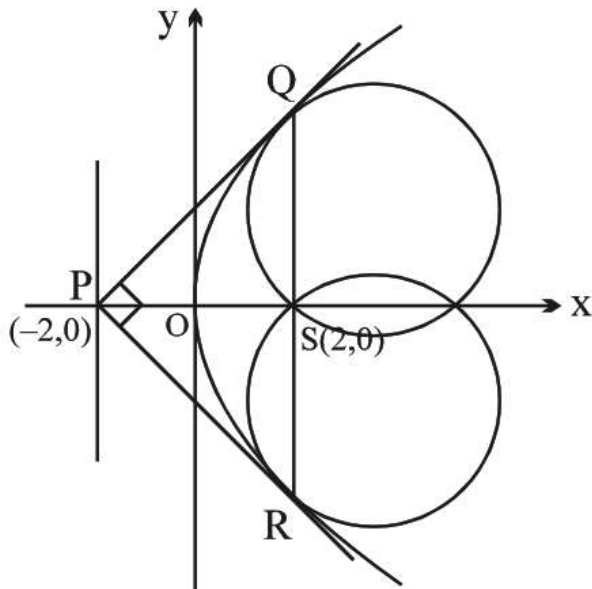
Now interpret.

- Q17. Through a point  $P(-2, 0)$ , tangents  $PQ$  and  $PR$  are drawn to the parabola  $y^2 = 8x$ . Two circles each passing through the focus of the parabola and one touching at  $Q$  and other at  $R$  are drawn. Which of the following point(s) with respect to the triangle  $PQR$  lie(s) on the radical axis of the two circles?

(A) centroid (B) orthocentre (C) incentre (D) circumcentre

Ans. A, B, C, D

Sol.



$(-2, 0)$  is the foot of directrix.

Hence Q and R are the extremities of the latus rectum and angle  $\angle QPR = 90^\circ$  with  $\triangle PQR$  as right isosceles.

Hence by symmetric the common chord of the two circles will be the x-axis which will be the median, altitude, angle bisector and also the perpendicular bisector.

Hence centroid, orthocentre, incentre and circumcentre all will lie on it.

Q18. If a number of ellipse be described having the same major axis  $2a$  but a variable minor axis then the tangents at the ends of their latera recta pass through fixed points which can be

- (A)  $(0, a)$     (B)  $(0, 0)$     (C)  $(0, -a)$     (D)  $(a, a)$

Ans. C, A

Sol.  $e$  is a variable quantity

$$\frac{xae}{a^2} + \frac{yb^2}{ab^2} = 1 \Rightarrow ex + y = a \Rightarrow y - a + ex = 0$$

it passes through  $(0, a)$ .

|||ly other point is  $(0, -a)$

Q19. The locus of the point of intersection of those normals to the parabola  $x^2 = 8y$  which are at right angles to each other, is a parabola. Which of the following hold(s) good in respect of the locus?

- (A) Length of the latus rectum is 2.    (B) Coordinates of focus are  $\left(0, \frac{11}{2}\right)$

(C) Equation of a director circle is  $2y - 11 = 0$

(D) Equation of axis of symmetry is  $y = 0$ .

Ans. C, A

Sol. Locus is  $x^2 - 2y + 12 = 0$

Q20. Solutions of the differential equation  $(1 - x^2) + xy = ax$  where  $a \in \mathbb{R}$ , is

- (A) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinates axes.  
(B) centre of the conic is  $(0, a)$  (C) length of one of the principal axes is 1.  
(D) length of one of the principal axes is equal to 2.

Ans. A, B, D

Sol.  $\frac{dy}{dx} + \frac{x}{1-x^2}y = \frac{ax}{1-x^2}$

$$\text{I.F. } e^{\int \frac{x}{1-x^2} dx} = e^{\frac{1}{2} \log|1-x^2|} = \frac{1}{\sqrt{|1-x^2|}}$$

$$\frac{y}{\sqrt{|1-x^2|}} = a \int \frac{x dx}{|1-x^2|^{\frac{3}{2}}} + C$$

$$\text{let } |1-x^2| = v^2; \quad -2x dx = 2v dv; \quad x dx = -v dv$$

$$\text{hence } \frac{y}{v} = -a \int \frac{v dv}{v^3} = +\frac{a}{v} + C$$

$$y = a + Cv$$

$$y = a + C\sqrt{|1-x^2|}$$

$$(y-a)^2 = C^2 |1-x^2| = C^2(1-x^2) \text{ or } C^2(x^2-1)$$

$$(y-a)^2 + C^2 x^2 = C^2 \text{ or } (y-a)^2 - C^2 x^2 = -C^2$$

$$\frac{(y-a)^2}{C^2} + \frac{x^2}{1} = 1 \text{ or } \frac{(y-a)^2}{C^2} - \frac{x^2}{1} = -1 \Rightarrow \text{centre } (0, a)$$